

WEEK – 02

Quick Sort:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <time.h>
```

```
#define MAX 1000000
```

```
void swap(int *a, int *b) {
```

```
    int temp = *a;
```

```
    *a = *b;
```

```
    *b = temp;
```

```
}
```

```
int partition(int arr[], int low, int high) {
```

```
    int pivot = arr[high];
```

```
    int i = low - 1;
```

```
    for (int j = low; j < high; j++) {
```

```
        if (arr[j] < pivot) {
```

```
            i++;
```

```
            swap(&arr[i], &arr[j]);
```

```
        }
```

```
    }
```

```
    swap(&arr[i + 1], &arr[high]);
```

```
    return i + 1;
```

```
}
```

```
void quickSort(int arr[], int low, int high) {
```

```
    if (low < high) {
```

```
        int pi = partition(arr, low, high);
```

```
        quickSort(arr, low, pi - 1);
```

```

        quickSort(arr, pi + 1, high);
    }
}

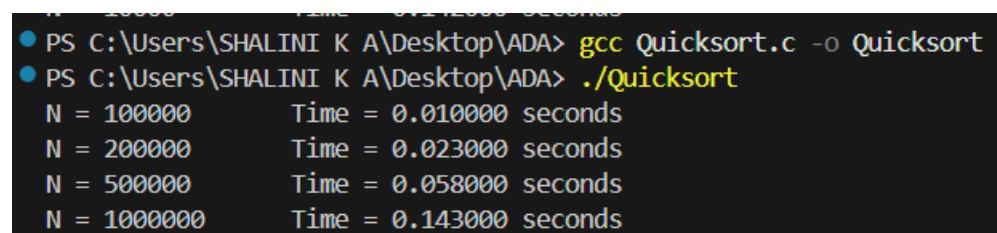
int main() {
    int sizes[] = {100000, 200000, 500000, 1000000};
    int *arr;

    srand(time(0));

    for (int i = 0; i < 4; i++) {
        int n = sizes[i];
        arr = (int*)malloc(n * sizeof(int));
        for (int j = 0; j < n; j++) {
            arr[j] = rand();
        }
        clock_t start = clock();
        quickSort(arr, 0, n - 1);
        clock_t end = clock();
        double time_taken = (double)(end - start) / CLOCKS_PER_SEC;
        printf("N = %d \t Time = %.6f seconds\n", n, time_taken);
        free(arr);
    }
    return 0;
}

```

Output:



```

PS C:\Users\SHALINI K A\Desktop\ADA> gcc Quicksort.c -o Quicksort
PS C:\Users\SHALINI K A\Desktop\ADA> ./Quicksort
N = 100000      Time = 0.010000 seconds
N = 200000      Time = 0.023000 seconds
N = 500000      Time = 0.058000 seconds
N = 1000000     Time = 0.143000 seconds

```